

ImagineUAV: Aerial Vision-Language Navigation via World-Action Modeling and Kinodynamic Planning

Xuchen Liu, Jiawei Huang, Shihao Xia, Bingxi Liu, Jinqiang Cui and Jiankun Yang

Abstract—Vision-language navigation (VLN) for UAVs demands grounding free-form instructions into 6-DoF flight under partial observability. While Vision-Language-Action (VLA) models excel at semantic reasoning, they suffer from brittleness due to geometric inconsistency and dynamics mismatch. To address this, we propose ImagineUAV, an imagination-driven framework leveraging cascaded world-action modeling. Instead of direct regression, ImagineUAV employs a latent video diffusion model to generate instruction-conditioned future observations—explicitly “imagining” environmental evolution—from which 6-DoF motions are inferred via an action extractor. A kinodynamic planner then refines these estimates into collision-free trajectories. Additionally, a step-distilled inference pipeline ensures real-time execution. With only 1.3B parameters, ImagineUAV outperforms prior VLN and VLA baselines on benchmarks and real-world flights, validating the practicality of imagination-driven aerial navigation. Video demo: <https://www.youtube.com/watch?v=Ng1alP0yhcl>.

Index Terms—vision-language navigation, world model, unmanned aerial vehicle, embodied AI

I. INTRODUCTION

VISION-language navigation (VLN) investigates how an embodied agent executes free-form natural-language instructions in unseen environments. Unlike conventional autonomous navigation, VLN is fundamentally instruction-centric and perception-driven: the agent must continuously align linguistic intent with streaming egocentric observations and infer actions under partial observability [1]. In aerial settings, the difficulty is further amplified by six-degree-of-freedom (6-DoF) motion, rapid viewpoint variation, payload-constrained sensing, and strict safety margins in cluttered 3D spaces. Small grounding or geometric estimation errors can escalate quickly into unstable or unsafe flight behavior, especially when operators issue high-level commands (e.g., “fly through the corridor and rise above the shelf”) instead of dense waypoint trajectories.

With the rapid rise of language-based foundation models, especially large language models (LLMs) and vision-language models (VLMs), an increasing number of VLN methods have adopted them as high-level planners, yielding notable gains in instruction understanding, semantic grounding, and

subgoal decomposition. However, this planning strength fails to transfer to low-level UAV control: 6-DoF flight demands geometric consistency and dynamics awareness that transcend semantic reasoning. VLA models [2] attempt to bridge this by introducing action tokens into the transformer backbone. However, the LLM/VLM backbone remains optimized for semantic prediction rather than embodied dynamics, making learned action heads brittle under out-of-domain shifts [3]. Hence, a core challenge persists: how to produce executable low-level commands from language-level intent while maintaining geometric consistency under real-world UAV dynamics.

In parallel, video-generation-based world models have progressed rapidly, showing a strong ability to predict temporally coherent future observations conditioned on context or action [4]. These advances motivate a different route: instead of forcing language models to output controls directly, one can ask a world model to *imagine what the agent would see if it follows the instruction* [5]. If imagination is sufficiently realistic and instruction-aligned, the generated video implicitly contains a future route, obstacle interaction cues, and motion scale. A geometric backend can then extract executable trajectories from this visual future.

Based on this insight, we propose *ImagineUAV*, an imagination-driven UAV vision-language navigation framework composed of three tightly coupled modules: (i) an instruction-conditioned world model that predicts future egocentric observations over a planning horizon; (ii) a geometry-aware action module implemented through learning-based visual odometry that extracts relative 6-DoF poses from imagined frames; and (iii) a kinodynamic planner that converts the extracted poses into collision-free, dynamically feasible trajectories. Specifically, through instruction-conditioned training, we enable the video generation model to accept VLN instructions and produce embodied future observation sequences covering diverse 6-DoF aerial motion patterns, which serve as a bridge between semantics and geometry. Language contributes task intent, and the world model materializes intent into controllable pixel-level futures. Unlike videos induced by real physical motion, imagined videos can exhibit spatio-temporal inconsistencies (e.g., object drifting and local temporal jitter); therefore, we carefully design a learning-based visual-odometry action extractor that is robust to such artifacts and still recovers reliable pose series at execution time. Since the raw poses may violate dynamical constraints and lack explicit collision checking, a kinodynamic planner refines them into smooth, collision-free, dynamically feasible trajectories for reliable flight execution. Finally, we deploy the system on a real UAV platform, where step-distilled inference

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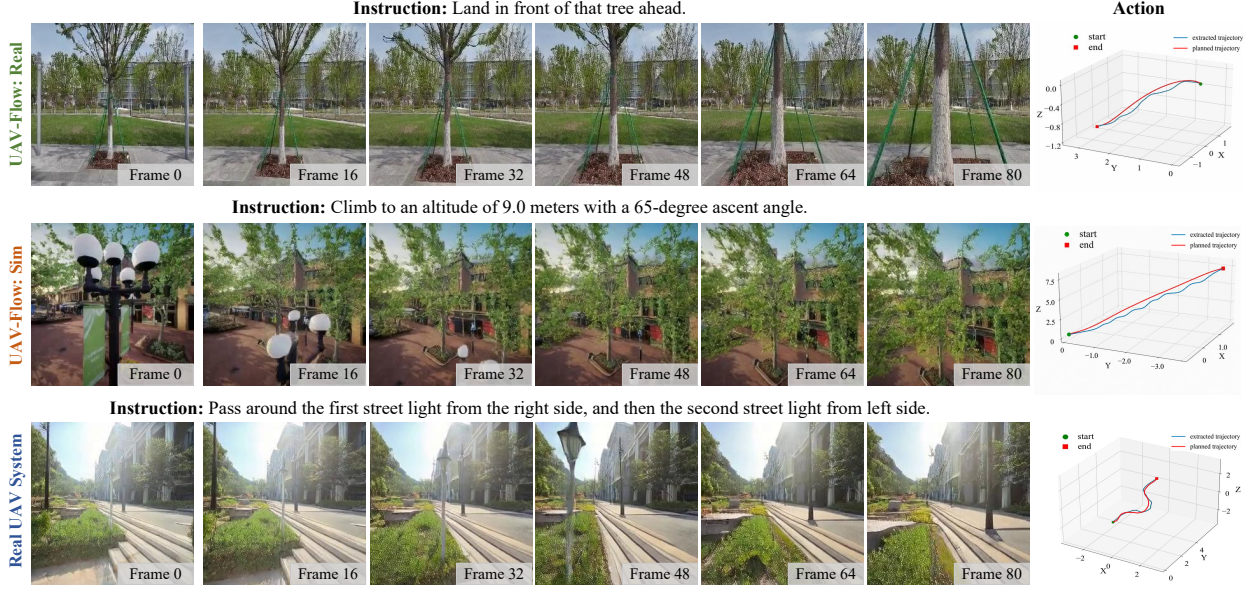


Fig. 1. Imagination-guided UAV navigation across diverse environments. Each row illustrates the instruction-conditioned generated observations alongside the corresponding flight execution results. The top and middle rows present UAV-Flow [6] cases, while the bottom row shows a real UAV system case.

provides practical planning latency and the onboard flight stack executes the generated waypoint references. Experiments validate the feasibility of our method, as illustrated in Fig. 1.

The contributions of this paper are summarized as follows:

- We propose an instruction-conditioned latent video diffusion world model for 6-DoF aerial VLN, which explicitly imagines future egocentric observations conditioned on language instructions and grounds motion intent through masked camera-control training.
- We design a spatio-temporal action extractor that robustly infers executable relative motions from imagined videos. A downstream kinodynamic planner further converts these estimates into collision-free and dynamically feasible trajectories, mitigating the brittleness of direct action execution.
- We integrate the proposed modules into a UAV navigation system with step-distilled inference for low-latency world-model generation. Extensive simulation experiments on the UAV-Flow benchmark [6] demonstrate state-of-the-art performance (70.9% success rate), while real-world flight tests consistently validate the effectiveness and practicality of the proposed framework.

II. RELATED WORK

A. Vision-Language Navigation and Action Learning

VLN has progressed from discrete graph-based benchmarks [7] to continuous-control settings [8] that require agents to execute low-level motion commands under noisy egocentric perception, better reflecting real-world robotic constraints. More recently, foundation-model-based approaches treat VLN as a high-level planning problem [9], leveraging LLMs and VLMs for instruction decomposition while delegating low-level control to separate modules. A popular paradigm, VLA models [10]–[12], unifies perception, reasoning, and control by

co-fine-tuning multimodal foundation models on large-scale robotic trajectories [3], predicting end-to-end action tokens directly from visual and linguistic inputs. In aerial settings, VLA frameworks [6], [13] have been proposed to regress 6-DoF control commands from raw pixels. However, direct action regression remains fundamentally limited by the lack of explicit future-state reasoning: without anticipating how the scene evolves, VLA backbones—optimized primarily for semantic prediction—struggle with geometric inconsistency, dynamics mismatch, and compounding errors over long horizons. This motivates an alternative paradigm that reasons over imagined future observations before committing to executable actions.

B. World Action Models

Recent advances in generative video modeling have enabled powerful predictive models of future observations. By combining these advances with the objective of modeling environment dynamics for planning and control [14], World Action Models (WAMs) [15] jointly model future observations and actions within a unified generative framework. These approaches have diverged along three complementary routes. Latent-space methods [16] predict compressed future states in a learned embedding and decode actions directly, offering computationally efficient planning but without explicit visual output. Pixel-level methods [5], [17] generate full future visual observations, preserving scene structure and motion scale for human-interpretable inspection. Hybrid approaches such as MoWM [18] combine both paradigms, reporting that latent representations better capture dynamics while pixel features preserve geometric details necessary for precise action decoding. Despite this progress, existing WAM research has primarily focused on manipulation, policy learning, and direct action decoding in limited action spaces. The integration of instruction-conditioned visual imagination with executable

action generation and motion planning remains largely unexplored for aerial vision-language navigation. Moreover, existing approaches [19] often rely on large external generative models and incur substantial inference costs without domain-specific adaptation.

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